

Lemma 3.2. *Let $k > 0$ and let $0 < \epsilon < 1/4$. Assume that*

$$(3.1) \quad \mathbb{P}[g(X) \geq k] \leq \epsilon.$$

Then

$$\mathbb{E}[X_{f(Y)}Y_{g(X)}] \leq \frac{1}{2} - (1 - 4\epsilon)2^{-k+1}.$$

Proof. Recall our notation $W_{f,g} = \mathbb{E}[X_{f(Y)}Y_{g(X)}]$. Since X is independent of Y , since X and $-X$ have the same distribution, and since Y and $-Y$ have the same distribution, (X, Y) has the same joint distribution as $(X, -Y)$, $(-X, Y)$ or $(-X, -Y)$. Hence

$$W_{f,g} = \mathbb{E}[-X_{f(-Y)}Y_{g(X)}] = \mathbb{E}[-X_{f(Y)}Y_{g(-X)}] = \mathbb{E}[X_{f(-Y)}Y_{g(-X)}].$$

Summing these and rearranging yields

$$4W_{f,g} = \mathbb{E}[(X_{f(Y)} - X_{f(-Y)})(Y_{g(X)} - Y_{g(-X)})].$$

Let B_Y be the event that $Y_1 = Y_2 = \dots = Y_{k-1}$, and note that $\mathbb{P}[B_Y] = 2^{-(k-2)}$. Write

$$(3.2) \quad \begin{aligned} 4W_{f,g} = & \mathbb{E}[(X_{f(Y)} - X_{f(-Y)})(Y_{g(X)} - Y_{g(-X)})\mathbb{1}_{\{B_Y\}}] \\ & + \mathbb{E}[(X_{f(Y)} - X_{f(-Y)})(Y_{g(X)} - Y_{g(-X)})\mathbb{1}_{\{B_Y^c\}}]. \end{aligned}$$

Consider the first term on the right hand side. Since $\mathbb{P}[g(-X) \geq k] = \mathbb{P}[g(X) \geq k] \leq \epsilon$, we have that $\mathbb{P}[g(X) \geq k \text{ or } g(-X) \geq k] \leq 2\epsilon$, by the union bound. Now, if $g(X) < k$ and $g(-X) < k$ then, under the event B_Y , $Y_{g(X)} = Y_{g(-X)}$. Hence

$$(Y_{g(X)} - Y_{g(-X)}) \cdot \mathbb{1}_{\{B_Y\}} = (Y_{g(X)} - Y_{g(-X)}) \cdot \mathbb{1}_{\{g(X) \geq k \text{ or } g(-X) \geq k\}} \cdot \mathbb{1}_{\{B_Y\}}.$$

We thus have by independence of X and Y that

$$\begin{aligned} & \mathbb{E}[(X_{f(Y)} - X_{f(-Y)})(Y_{g(X)} - Y_{g(-X)})\mathbb{1}_{\{B_Y\}}] \\ &= \mathbb{E}[(X_{f(Y)} - X_{f(-Y)}) \cdot (Y_{g(X)} - Y_{g(-X)}) \cdot \mathbb{1}_{\{g(X) \geq k \text{ or } g(-X) \geq k\}} \cdot \mathbb{1}_{\{B_Y\}}] \\ &\leq \mathbb{E}[4 \cdot \mathbb{1}_{\{g(X) \geq k \text{ or } g(-X) \geq k\}} \cdot \mathbb{1}_{\{B_Y\}}] \\ &= 4\mathbb{P}[g(X) \geq k \text{ or } g(-X) \geq k, B_Y] \\ &= 4\mathbb{P}[g(X) \geq k \text{ or } g(-X) \geq k]\mathbb{P}[B_Y] \\ &\leq 8\epsilon 2^{-(k-2)}. \end{aligned}$$

Consider now the second term on the right hand side of (3.2). Noting that $\mathbb{P}[X_{f(Y)} = X_{f(-Y)} | Y = y] \geq 1/2$ for any y , we get that

$$\mathbb{E}[(X_{f(Y)} - X_{f(-Y)})(Y_{g(X)} - Y_{g(-X)})\mathbb{1}_{\{B_Y^c\}}] \leq 2\mathbb{P}[B_Y^c] = 2(1 - 2^{-(k-2)}).$$

Substituting these two bounds back into (3.2) and dividing by 4, we get

$$W_{f,g} \leq \frac{1}{4} \left(8\epsilon 2^{-(k-2)} + 2(1 - 2^{-(k-2)}) \right) = \frac{1}{2} - (1 - 4\epsilon)2^{-k+1}.$$

□